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Impact of climate change adaptation on paddy yield in dry zone Sri Lanka: A case study using agricultural production systems simulator (APSIM) model

Eranga M Wimalasiri *, A D Ampitiyawatta, P K Dissanayake[1] and A S Karunaratne

Department of Export Agriculture, Faculty of Agricultural Sciences, Sabaragamuwa University of Sri Lanka, Belihuloya, 70140, Sri Lanka
*Corresponding author: eranga@agri.sab.ac.lk

Abstract. Climate change in terms of increased temperature and variation of rainfall patterns is evident in Sri Lanka. Being a country with an agro-based economy, the impact of the changes of the climate negatively affects crop production and a possesses threat to food security. Using a crop modelling approach, the impact of climate change adaptation on paddy, which is the major crop in the country was assessed in Thanamalwila (6.43ºN, 81.09ºE), Sri Lanka. The calibrated and validated Agricultural Production Systems Simulator (APSIM) model was used for yield simulations. Bias-corrected future (2040-2069) climate data were obtained from 3 general circulation models; BNU-ESM, CSIRO-Mk3-6-0, and GFDL-ESM2M under the RCP4.5 scenario. Short season (BG300) and long season (BG358) cultivar, late and early planting (both by two weeks) were used as adaptation methods. Average paddy yield under baseline (1980-2009) climate for BG300 and BG358 were 3711 and 4524 Kg ha^{-1}, respectively. In BG300, the average future yield decreased in all 3 GCMs while in BG358 (long season), the average yield increased under 2 GCMs except CSIRO-Mk3-6-0. On average, delayed planting by 14 days as an adaptation measure increased the yield of BG300 (2.43%), the short season cultivar while in BG358, the average yield decreased by 3.55%. Except for CSIRO-Mk3-6-0 (in BG358), paddy yield in both cultivars decreased on early planting. This simple crop modelling exercise has shown the possibility of selection of cultivar and planting dates as climate change adaptation strategies. Further, field experiments and detailed crop model simulations are essential to validate the findings.

Keywords: APSIM, climate change adaptation, crop modelling, food security, rainfed agriculture

1. Introduction

Climate change is one of the major constraints to crop production in many parts of the world including Sri Lanka. According to previous studies, climate change is evident in Sri Lanka in terms of rainfall variability, increased temperature, and a higher frequency of extreme climatic events [1] [2]. With the climate changes, the yields of key crops in tropical regions such as paddy and maize are expected to decline [3] [4]. Due to climate change, it is expected that paddy yield will be decreased by 27.5% in China for the 2041-2070 period [5]. Sri Lanka is a tropical country on which rainfed agriculture is prominent [6] and around 34% of the total cultivated area of the country is under rice cultivation. Therefore, being a country with an agricultural-based economy, the impact of the changes of the climate negatively affects crop production and a possesses threat to the food security in Sri Lanka.

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Different approaches that include empirical relationships and crop models have been used to study the impact of climate change and climate adaptation worldwide. Out of them, crop models act as one of the fast and reliable tools to assess the climate change impact and adaptation strategies under diverse agroecological conditions and crop varieties [7]. The Agricultural Production Systems Simulator (APSIM) is one of the crop modelling platforms to simulate crop production systems under climatic and management options [8]. Previously, it has been successfully used to simulate rice production in Sri Lanka [9] [10] [11].

Adaptation options are one of the best solutions to minimize crop yield losses due to climate change [1]. Farmers in Sri Lanka have their own adaptation measures to avoid unfavourable weather conditions and yield losses [1] [2]. However, they were not studied in detail using crop models. Therefore, the objective of this study is to assess the impact of climate change adaptation on paddy using the crop modelling approach. Paddy yield will be simulated using the Agricultural Production Systems Simulator (APSIM) model.

2. Materials and method

2.1. Study area

Thanamalwila (6.43°N, 81.09°E) which is located in the dry zone of Sri Lanka was selected as the study location. The annual average rainfall of the area is less than 1500 mm. A bi-modal pattern of rainfall can be observed in the area on which two peaks in March-April and October-November lead to the onset of two growing seasons.

2.2. Climate data

Observed daily minimum and maximum temperature and rainfall data for the 1980-2009 period were collected from the Department of Meteorology Sri Lanka. Due to the unavailability of observed solar radiation, the data were obtained from the NASA POWER database [12]. Data were quality checked for errors and missing data.

Bias-corrected daily minimum and maximum temperature, rainfall, and solar radiation for mid-century climate (2040-2069 period) were obtained for Representative Concentration Pathway (RCP) 4.5 scenario. Data were downscaled for 3 General Circulation Models (GCMs) as; BNU-ESM (GCM1) of College of Global Change and Earth System Science, Beijing Normal University, China, CSIRO-Mk3-6-0 (GCM2) of Commonwealth Scientific and Industrial Research Organization, and the Queensland Climate Change Centre of Excellence, Australia and GFDL-ESM2M (GCM3) of Geophysical Fluid Dynamics Laboratory, USA. The data were downscaled from the Consultative Group for International Agricultural Research (CGIAR) Research Program on Climate Change, Agriculture and Food Security (CCAFS) database (http://ccafs-climate.org/).

2.3. Model description

The Oryza module of Agricultural Production Systems Simulator (APSIM - Version 7.10) [8] was used to simulate paddy growth and development. Being a process-based crop model, APSIM can simulate several crops, climate, and soil-related processes. The model requires daily rainfall, minimum and maximum temperatures, and solar radiation as climate data.

2.4. Climate change adaptation options

Two types of adaptation options were selected as cultivar and planting date. BG300, a short duration (3 months) cultivar, and BG358, a long duration (3 and half months) cultivar were selected. These two cultivars are prominent cultivars across Sri Lanka. Two planting dates were used as; late planting (2 weeks delayed planting) and early planting (2 weeks earlier planting).

2.5. Yield simulation

Pre-calibrated and validated APSIM Oryza model was used in this study [10]. Published genetic coefficients of paddy cultivars were used in simulations. All other parameters of the APSIM Oryza model were set as default. The planting date was set as 10th November. All other crop management data (plant densities, fertilizer applications, etc.) were collected from Wimalasiri et al. [9]. Paddy yield was simulated for both past/ baseline (1980-2009) and mid-century (2040-2069) climates.

3. Results and discussion

3.1. Past and future climate

The observed annual average of maximum (Tmax) and minimum (Tmin) temperatures of Thanamalwila were 32.1 ºC and 23.3 ºC, respectively. The average annual rainfall for the baseline period was 1121 mm. In all 3 GCMs considered, both Tmax and Tmin increased during the 2040-2069 period on which the highest Tmax and Tmin increments were reported from GFDL-ESM2M (1.3 ºC) and CSIRO-Mk3-6-0 (1.7 ºC) respectively. The average annual rainfall increased by 129 mm in BNU-ESM and by 507 mm in CSIRO-Mk3-6-0 while decreasing by 76 mm in GFDL-ESM2M. The variation of rainfall and minimum and maximum temperatures in both past and future climates are shown in Figure 1.

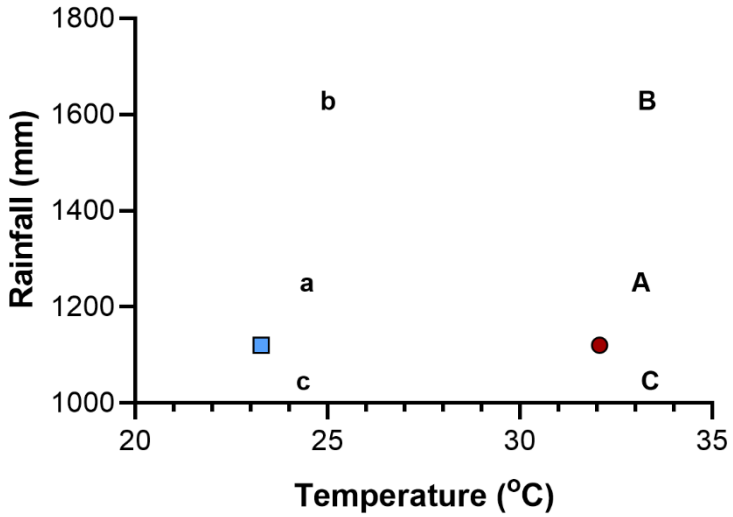


Figure 1. Variation of minimum and maximum temperatures and rainfall during 1980-2009 (symbols) and 2040-2069 (letters) period. In symbols, the square represents minimum temperature while the circle represents maximum temperature. Letters denote 3 GCMs while lower and uppercase letters represent minimum and maximum temperatures

3.2. Paddy yield and yield change in future climate

The average paddy yield for both past and future climates is presented in Table 1. Comparatively, a higher yield was reported from the long duration cultivar (BG358) than the short duration cultivar (BG300).

Table 1. Average paddy yield of two cultivars during past and future climates

Climate	Yield (kg ha^{-1})	
	BG300	BG358
Past climate	3711±255	4524±327
Future climate - BNU-ESM	3646±218	4617±323
Future climate - CSIRO-Mk3-6-0	3467±245	4455±361
Future climate - GFDL-ESM2M	3668±244	4575±335

When compared with the past yield, the average paddy yield decreased in all 3 GCMs in BG 300 with the highest reduction of 6.6% in CSIRO-Mk3-6-0. The average yield reduction of BG300 in BNU-ESM and GCM2 were 1.8 and 1.1% respectively. In the long duration cultivar (BG358), the paddy yield decreased by 1.5% in CSIRO-Mk3-6-0 while increased in 2 other GCMs by 2.1% (BNU-ESM) and 1.1% (GFDL-ESM2M). The results of this study agree with the previous findings on projected paddy yield for mid-century (2040–2070) in the Intermediate zone, Sri Lanka using other GCMs [10].

In contrast, some other studies in dry zone Sri Lanka have shown that the yield of BG300 cultivar is expected to increase in the year 2050 compared to the year 2017 [11]. The impact of climate change on crop yield depends on various factors that include location, season, and management practices. Therefore, detailed studies are needed on the performance of BG300 and BG358 in other agroecological zones in Sri Lanka.

3.3. Impact of adaptation options

The paddy yield change under adaptation measures is shown in Figure 2. The impact of late and early plating on the baseline climate was initially studied. According to the model simulated paddy yield, the late planting increases the yield of short duration cultivar (BG300) by 1.92% while decreases in the long duration cultivar (BG358) by 1.43% in the current climate. Early planting by 14 days decreased the yield in both cultivars (by 0.64 and 0.56% respectively).

However, in the future climate, the delayed planting by 14 days increased the yield in all 3 GCMs for short season cultivar on which the values ranged from 1.71% (GFDL-ESM2M) to 3.09% (CSIRO-Mk3-6-0). In contrast, the yield of long duration cultivar (BG358) decreased on delayed planting in all 3 GCMs with the highest reduction of 4.51% in CSIRO-Mk3-6-0. In early planting by 14 days, the yield decreased in all the GCMs for short duration cultivar. However, early planting increased yield in CSIRO-Mk3-6-0 (0.11%) while decreasing in two other GCMs in the long duration cultivar.

Two adaptation strategies (cultivar selection and change of planting date) used in this study showed different sensitivities under different future climate scenarios. However, detailed field experiments and yield simulations using crops models are needed to validate the findings. Other than these options, farmers in Sri Lanka follow different climate change adaptation strategies that can be categorized into land management, crop management, and irrigation management [1]. The sensitivity of other climate change adaptation strategies can be assessed using the crop modelling approach.

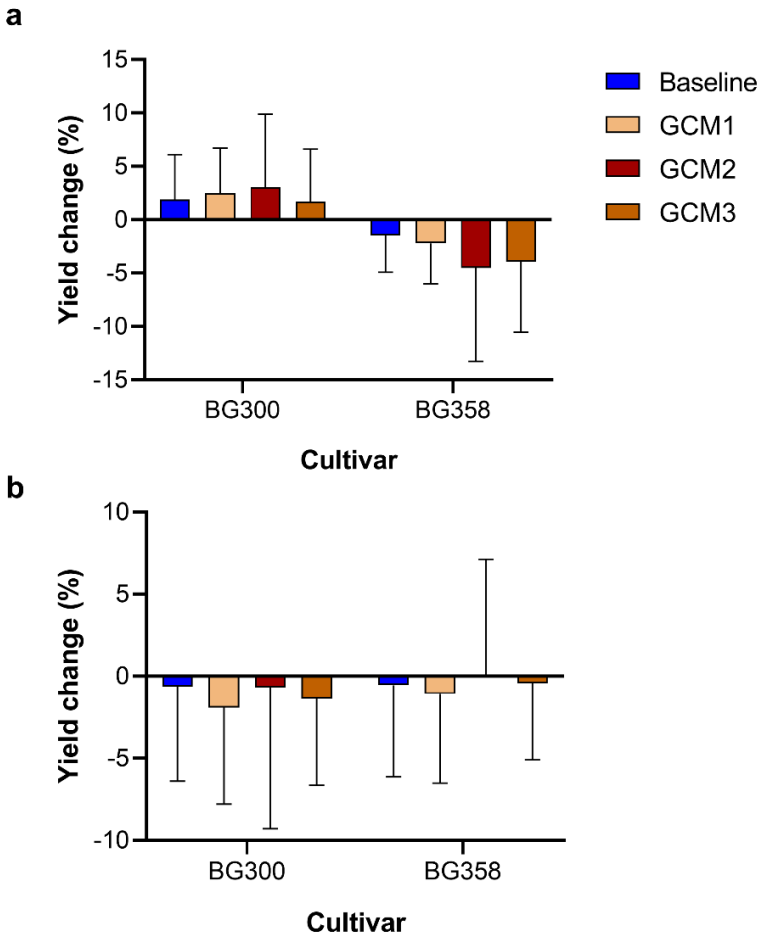


Figure 2. Variation of paddy yield change of baseline (1980-2009) and future (2040-2069) climate under late planting (a) and early planting (b)

4. Conclusions

Both minimum and maximum temperatures and rainfall (average) are expected to increase in Thanamalwila, Sri Lanka during the 2040-2069 period. The Agricultural Production Systems Simulator (APSIM) model has been successfully used to study the climate change adaption strategies in Sri Lanka. According to the model output, the short-duration rice cultivar (BG300) is at high risk than the long duration cultivar (BG358) in the mid-century (2040-2069) climate. This simple crop modelling exercise has shown that selection of cultivar and planting dates are possible climate change adaptation strategies in the dry zone, Sri Lanka.

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